KSC Diffusion: K-Space Cold Diffusion Architecture

Undersampling

Single-coil k-space K_0, mask M

$$y = M \odot K_0$$

R4 random column mask, centre fraction 0.08

Cold Diffusion Forward Process

(Training only)

Deterministic degradation $D_t(\cdot)$: removes unacquired columns only

$$K_0 \to K_1 \to \cdots \to K_T = y$$

$$x_t = \mathcal{F}^{-1} D_t(K_0)$$

$$t \sim \mathcal{U}\{0, \ldots, T\}, \text{ target } x_0 = \mathcal{F}^{-1} K_0$$

fixed random removal order per sample

Denoising Network

Training objective:

$$\mathbb{E}_{K_0, t} \| f_\theta(x_t, t) - x_0 \|_1$$

Timestep t → Sinusoidal embedding → MLP

Input x_t $(2 \times H \times W)$ → Time-conditioned 2D U-Net (below)

Predicted clean image $\hat{x}_0 = f_\theta(x_t, t)$ $(2 \times H \times W)$, real and imaginary parts

no noise is added or predicted

Reverse Cold Diffusion

(Inference only)

start: $x_T = \mathcal{F}^{-1} y$ (zero-filled)

$$\hat{k}_t = \mathcal{F} f_\theta(x_t, t)$$

$$x_{t-1} = x_t - \mathcal{F}^{-1} D_t(\hat{k}_t) + \mathcal{F}^{-1} D_{t-1}(\hat{k}_t)$$

repeat for $t = T, \ldots, 1$

acquired columns stay equal to y

Reconstruction Output

x_0 after the last reverse step → $|\cdot|$ magnitude → Reconstructed image

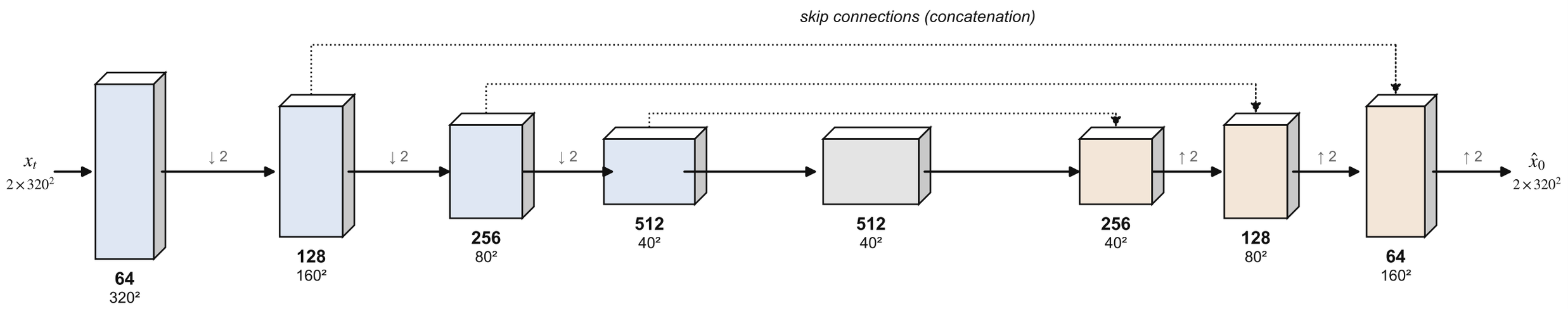
Diffusion-time configuration

$T_{\max} = 1000$ degradation levels; a configuration T traverses T evenly spaced levels from $t = T_{\max}$ to 0 with the same network

fixed-step sampler: NFE $= T$

same network for every configuration

Time-conditioned 2D U-Net (56.6 M parameters)



Encoder: 2 × ConvNeXt block + linear attention per level

Bottleneck: ConvNeXt + attention + ConvNeXt

Decoder: skip concatenated, 2 × ConvNeXt block + linear attention, then ↑ 2; final ConvNeXt + 1×1 conv

ConvNeXt block: depthwise 7×7 conv + time embedding, LayerNorm, 3×3 conv, GELU, 3×3 conv, residual. ↓ 2: 4×4 stride-2 conv; ↑ 2: 4×4 stride-2 transposed conv.